

can't revert to the plaintext by replacing the letters in the ciphertext by letters two place back. You'll have to replace the letters by the same number of positions backwards.

Symmetric ciphers are named as such because both sides of the process (encryption and decryption) utilize the same key. Symmetric ciphers are also known as 'private key cryptography'.

Asymmetric Ciphers

Asymmetric ciphers are usually known as public key cryptography and are well known as 'public key cryptography' algorithms. In a public key cryptography method, the key which is used to encrypt plaintext to ciphertext can't be used to decrypt the ciphertext back into plaintext. The two keys involved are called as public key and private key.

Public Key: Public keys are used to encrypt plaintext. They're named as public keys because they can be made public.

Private Key: Private keys are used to decrypt ciphertext back to plaintext. They're named as such because they're meant to be private.

Both these keys are related to each other using a mathematical relationship in a way that if you're given one key, you wouldn't be able to derive the second key. This allows the public key to be public. Anyone can have it and send you a message encrypted with the public key but no one can decrypt an encrypted message being sent to you by another person.

Let's consider a hypothetical situation. Agent Archer wants to send a secret message to his boss, Jim; so he asks Jim for his public key. Jim creates a public-private key pair and sends the public key to Archer. A spy named Cyril uncovers the public key! Agent Archer then encrypts the message and sends it to his boss. The spy also intercepts the encrypted message. In the meanwhile, Jim has also received the encrypted message and he uses his private key to decrypt the message. Cyril is still trying to figure out how to decrypt the message using the key!

This property of asymmetric algorithms makes them indispensable in the modern era. They allow to freely send the key as well as the message! Since the key for decryption is separate from what is being used for encryption, it makes the communication safe!

However asymmetric algorithms are usually much more costly on the CPU than symmetric algorithms. This brings up an interesting usage of public key algorithm (asymmetric algorithm) whereby it is only used to transfer the key of a symmetric algorithm. This allows the key to be trans-